0.0350 mm 105.7 mg 9700 mL

Name: ______________________________
Period: ________ Date: ________________

Significant Figures & Precision

Main Idea	Notes

Uncertainty of Measurement

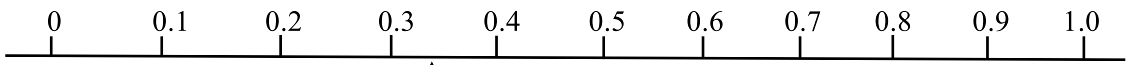
How "far out" can a measurement be made?

No measurement instrument can measure to ______________________________ A standard system of measurement is established that exactly shows the ________________ of the instrument used to make the measurement.

Measurement are made to ________________________ where the measuring device is marked.

The precision of a measurement (______________________________) is indicated by the number of ______________ recorded.

Device is <u>marked</u> to the **ONES** place. <u>Read</u> to the **TENTHS** place.



Device is <u>marked</u> to the **TENTHS** place. <u>Read</u> to the **HUNDRETHS**.

Device is <u>marked</u> to the **TENs** place. <u>Read</u> to the **ONES** place.

By reading the measurement, you can envision the markings on the ____________________ that was used. **56.7 cm** The measurement instrument had marks to the ______________ place.

97.755 g __________________________ place.

Precision

	Precision	Smallest Mark on Measuring Instrument
12.050 cm	________________	________________
5.15 mL	________________	________________
0.0070 g	________________	________________
72.325 s	________________	________________
0.900 mg	________________	________________
35.0 cm	________________	________________

175,000 cm ________________________

3,400 km ________________________

15,000 ppm ________________________

Precision is to the last ______________ digit.

What if we need to show that the precision is to a place with a zero?

15,000 mm ________________________

15,000 mm ________________________

15,000 mm ________________________

Show precision for zeros on the left side of the decimal with a line over the zero.
Use a decimal if it's the last zero!

	Precision	Scientific Notation
15,$\bar{0}$00 mm	________________	________________
15,0$\bar{0}$0 mm	________________	________________

Scientific Notation ONLY contains the __________ digits & an ______________ digit in order to show precision.

Another way to show precision is to use scientific notation instead of a line.

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Significant Figures & Precision

Main Idea	Notes
Significant Figures	Significant figures (________________) are digits that are actually ______________ (certain digits) ending with the last digit that is ____________________). Sig Figs = all ________________ digits and one ____________________ digit. The purpose of sig figs is to ____________________________________ of the measurement.

Rules for Counting Sig Figs

1. All ____________________ digits are significant.

   2348 = ________ sig figs
   175,348 = _______ sig figs

2. **Imbedded zeros** __________________ count as sig figs.

   16.07 = ________ sig figs
   20,034 = ________ sig figs

This just leaves the zeros to deal with!

3. **Leading zeros** <u>DO NOT</u> count as ____________________.

   0.04863 = ________ sig figs
   0.001 = ________ sig figs

4. **Trailing zeros** are significant only if the number contains a ____________________.

   2.800 = _____ sig figs   23.0 = ________ sig figs
   245.150 = _____ sig figs

Placeholder zeros: zeros that must be present to give ___________ to the __________________.
(keeps digits in the right place value) Placeholder zeros ARE NOT significant!

1000= ________ sig figs

0.00570 = _________ sig figs

What about this measurement? 203,000 What if we need to show the precision to the hundreds place, or the tens place, or the ones place?

Trailing zeros in a number larger than 1, are significant ONLY if they are followed by a decimal, or come before a zero with a line over it (include zero with the line).

Scientific Notation ONLY contains significant figures!

203,000 =

203,000 =

203,000 =

Instead of putting a line over the last significant zero, you can use scientific notation to show all sig figs.

$800,\bar{0}00$ =

8.00×10^5 =

Practice!

How many sig figs are in each of the following measurements?

4.0070 m	________
27.50 L	________
7,002,800. g	________
15.10×10^3 cm	________
0.000670 m	________
1,200,000 μm	________
$800,0\bar{0}0$ mg	________

TIP! Start counting with the first NON-ZERO on the LEFT, and count to the right.

0.0205 mg	35,000 mg

Numbers larger than ONE: Start with first non-zero and count to the decimal, line, or to the end if decimal is present.

35,000. mg	$35,0\bar{0}0$ mg	35,000.15 mg

0.17 mm	________
0.0030 mL	________
0.0900 L	________
0.100 cm	________
0.00008 km	________
0.012300 g	________

24300 nm	________
$150,0\bar{0}0$ ppm	________
4700. g	________
49.0 km	________
23.070 mg	________
70.000 g	________
8.10×10^{-5} g	________

43.8 g	________
2550 cm	________
470 mL	________
5.4070×10^{-2} L	________
0.0075000 kg	________
$65,0\bar{0}0,000$ mg	________
940,000. ppm	________

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Significant Figures & Precision Name: ____

Homework Period: _______Date: _____________________________

True or False

___ 1. Significant figures are used to show the precision of a measurement.

___ 2. Placeholder zeros are always significant.

___ 3. A correctly recorded measurement ends with an estimated digit.

___ 4. Imbedded zeros are not significant.

___ 5. In scientific notation all of the digits in the base number are significant.

___ 6. Leading zeros are always significant.

___ 7. A line over a zero indicates that it is the last significant digit.

___ 8. Trailing zeros to the right of the decimal are always significant.

___ 9. The number 550,000. has 6 sig figs.

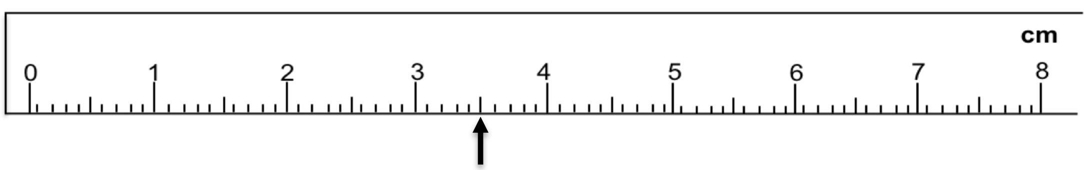
___ 10. The number 0.00570 has 6 sig figs.

Short Answer

11. Explain the importance of using significant digits. ____

____

____



12. Correctly record this measurement. ________________ How many sig figs does this measurement have? ______

13. Describe a placeholder zero. ____

____

How many sig figs?

Determine the number of sig figs for each of the following measurements.	Round the following measurements to the specified number of sig figs.
14) _____ 0.0120 m	19) ___________________________ 5,487,129 m to three sig figs
15) _____ 100.5 mL	20) ___________________________ 31,947.972 cm^2 to four sig figs
16) _____ 350. cm^2	21) ___________________________ 192.6739 cm^3 to five sig figs
17) _____ $10\bar{0}0$ kg	22) ___________________________ 786.09164 cm to two sig figs
18) _____ 0.0020700 dm	

Determine the number of sig figs for each of the following measurements.	Round the following measurements to the specified number of sig figs.
23) _____ 0.000150 L	27) ___________________________ 35.27 cm to three sig figs
24) _____ 998,870 cm^3	28) ___________________________ 1.35 m to two sig figs
25) _____ 1.000×10^4 kg	29) ___________________________ 78,000,000 nm to four sig figs
26) _____ 23,500 mg	30) ___________________________ 350.275 cm to three sig figs

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